



Spring DemoSat: Self-Orienting Payload

Critical Design Review

3.9.2026

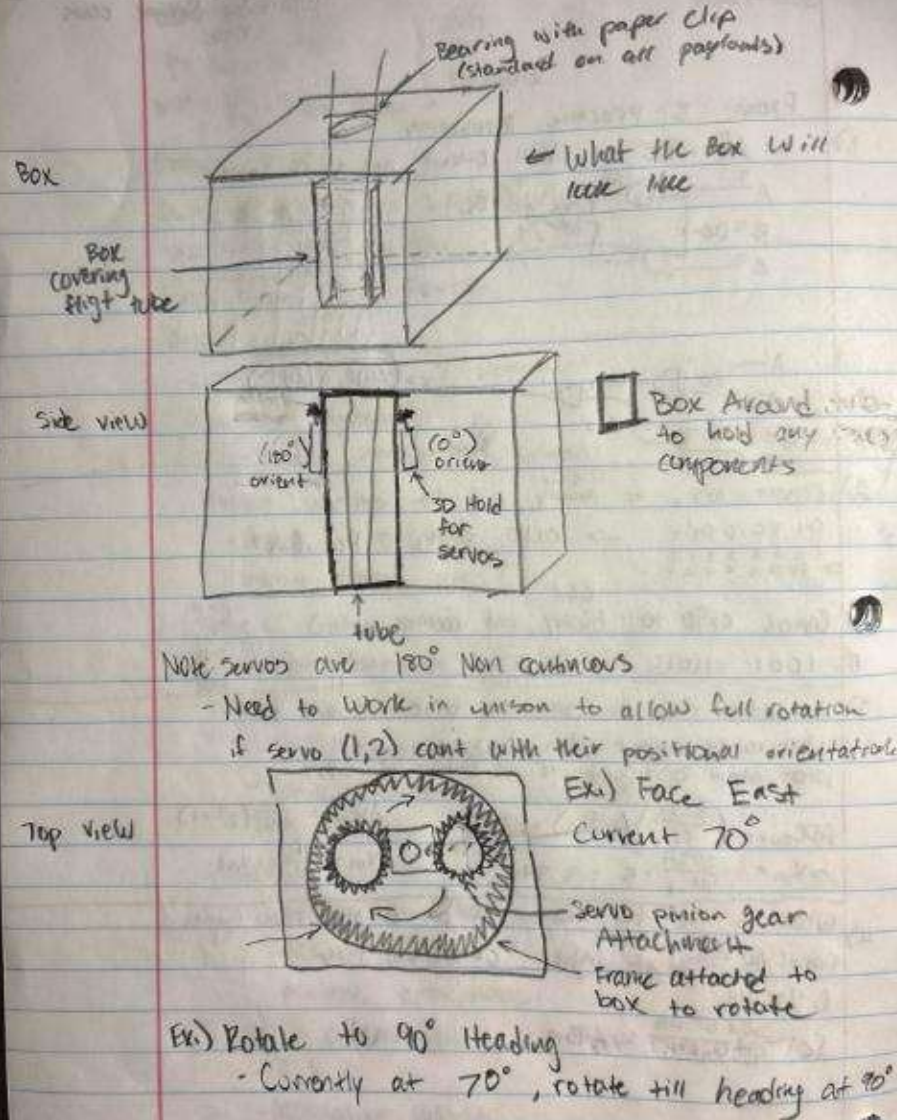
Mission Overview

- Design and build an atmospheric payload that will consistently be at a heading of 90° throughout the flight by orienting the payload around the fixed flight tube (not touched).
- Integrate photodiodes that will provide insight on voltage throughout the flight.
- Integrate environmental and payload health sensors that will produce data to interpret heading and ensuring that our payload withstands environmental conditions.
- Read and log data using integrated sensors and OpenLog for post flight analysis.
- Produce a professionally written paper and poster for the Spring Symposium.

Arduino Nano 33 BLE Sense Rev 2 – Hardware Overview

- Processor: ARM Cortex-M4 @ 64 MHz (32-bit, low-power)
- Memory: 256 KB Flash, 128 KB RAM
- Onboard Sensors (not used in this project)
 - Temp, humidity, barometric
 - IMU
 - Microphone
- Advantages
 - Low power consumption.
 - Lightweight and suitable for embedded flight system.
 - Enables an efficient deployment of TinyML models.

Design



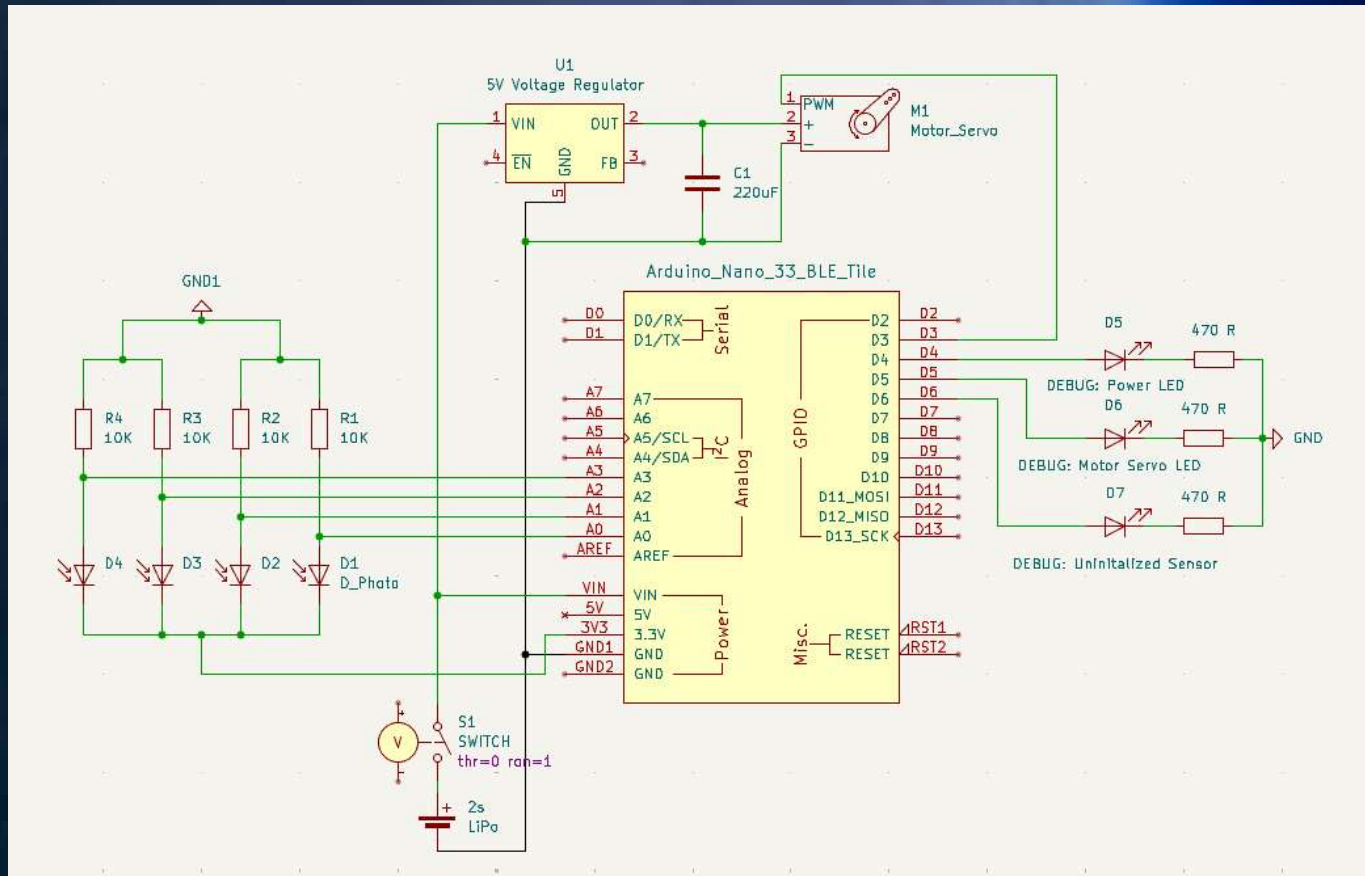
Mission Requirements

- Cost must be below \$500
- Temperature must withstand temperature down to -40 °C
- Payload weight must be below 800g
- Pressure resistant at high altitudes
- Battery must be able to withstand severe conditions
- Post-Flight code data analysis
- Motor needs to withstand the ability to rotate continuously.

Subsystem Requirements

- Power
 - Our power supply (2s LiPo), must be able to withstand extreme conditions while being able to supply our systems
 - We will need to power our: Switch, Arduino Nano 33 BLE Sense Rev2, and BME280, Servos, Heat Source, Solar Panel.
- Heat Source
 - 9V Battery with Large Resistor, Activated on specific altitude.

Electronics Schematic



Commands and Sensors

- When Switch == ON
 - Data collection starts
 - IMU
 - Feed this data to a workflow that will produce the heading required for the servo motors to rotate the payload at an angle to meet its criteria.
 - Save this angle and data throughout for post-flight accuracy.
 - Based on positional control.
 - BME280
 - Feed this data to read internal payload temperatures, if lower than expected, turn on heat source to preserve battery consumption.
 - From each data collected, saved onto an SD card.
 - I only intend to use one 8gb SD Card for this experiment
- When Switch == OFF
 - Data collection is NULL

Testing Plans

- Battery Capability
 - Environmental testing, Pressure testing
- Sensors
 - Drop test, Functionality, Environmental
- LEDs/Switch Functions
 - Whip, Drop, Functionality

Parts List

- 9V battery
- Sensors
 - OpenLog (SD card reader)
 - BME280 (Pressure & Temp sensor, built-in)
 - IMU (built-in)
- Photodiodes
- Solar Cell attached to East heading
- Microcontrollers
 - Arduino Nano 33 BLE Sense Rev2
- Box : Most likely will be in the ranges of 4x4 or 5x5 dimension.

Expected Results

- The payload keeps a relative stability orientation despite external disturbances like turbulence or tilt.
- The positional control system must detect the orientation changes and command the servo motors to correct them to head at its appropriate heading without affect internal components.
 - Must not affect flight tube but must be able to rotate around the fixed position using a pinion gear and track.
- The rotating mechanism must successfully adjust the payload to its proper orientation to reduce the error calculated post-flight.
- The system must prove an effectiveness and cooperability during extreme conditions to handle the goal of this project during testing.

Structure

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Conclusion

- We expect that our stabilization system successfully proves the ability to detect and correct its own orientation heading using sensor feedback integration for positional control.
- The servo-driven rotation mechanism effectively adjusts the payload under external disturbances.
- This project insights will hopefully lead into refining the control system and mechanic designs for a smoother motion and outlook during the summer 2026 flight.
 - Still plan to try to fly as a prototype to see how it performs.

Acknowledgements

- Professor Jones
- Professor Wikowsky
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